

In [5]: `!pip install -q google-genai`

```
In [7]: import json
from google import genai
from google.genai import types
from google.colab import userdata

# 1. Authenticate directly using your Colab Secret
api_key = userdata.get('GEMINI_API_KEY')

# 2. Initialize the Gemini Client
client = genai.Client(api_key=api_key)

# 3. Define the source passage and prompt
passage = """
MediPlus Pharmacy, located near City Hospital, recorded several customer
purchases during the third week of November.

On 18 November, Mr. Arun Kumar purchased Amoxicillin 500 mg and
Paracetamol 650 mg based on a doctor's prescription. The total bill
amounted to ₹785, which was paid using UPI. The medicines were issued
immediately, and the purchase status was marked as Completed.

Later that morning, Mrs. Kavitha Raman purchased Vitamin D3 Capsules,
Calcium Tablets, and Multivitamin Syrup costing ₹1,280 using a Credit
Card. Since these products were nutritional supplements, no prescription
was required.

During the afternoon, Mr. Mohammed Irfan, a diabetic patient, purchased
Insulin Pens, Blood Glucose Test Strips, and Lancets for ₹3,650. The
payment was completed using Net Banking, and the purchase was
successfully completed.

On 19 November, Mrs. Priya Senthil purchased a Digital Thermometer and a
Pulse Oximeter from the pharmacy's healthcare section. The total invoice
value was ₹2,150, and payment was made using a Debit Card.

Later that day, Mr. Vinoth Raj purchased Antiseptic Solution, Cotton
Rolls, Sterile Gauze Pads, and Adhesive Bandages worth ₹620. The payment
method was not mentioned, but the purchase was completed successfully.

On 20 November, Mrs. Deepika Anand visited the pharmacy with a pediatric
prescription and purchased Baby Cough Syrup, Infant Vitamin Drops, and
Baby Skin Cream. The invoice amount was ₹1,420, and payment was made
using Google Pay.

During the evening, Mr. Sanjay Verma purchased Blood Pressure Tablets,
Cholesterol Medication, and Vitamin B12 Tablets for ₹1,860, paying
through PhonePe.

On 21 November, Mrs. Nisha George purchased Face Wash, Moisturizer,
Sunscreen Lotion, and Hand Sanitizer costing ₹1,190 using Cash.

Later that afternoon, Mr. Ashok Babu purchased a Digital Blood Pressure
Monitor costing ₹2,950 using Credit Card. The device included a three-
year warranty and was delivered immediately.

Towards the end of the day, Mrs. Shalini Raj purchased Pain Relief
Spray, Muscle Relaxant Gel, and Pain Relief Tablets for ₹875. The
```

payment method was not specified, and the purchase was successfully completed.

"""

prompt = f"""

Objective: Analyze the pharmacy's daily sales records and extract all medicine purchase transactions into a structured JSON format.

Instructions:

- Extract all pharmacy sales transactions from the given passage.
- Return the extracted information strictly in valid JSON format as a list of objects.
- If any field is unavailable in the passage, return "Unknown".
- Bill amounts should contain only numeric values without the currency symbol (₹).
- Classify every medicine purchase strictly into one of the categories listed below.
- Do not generate explanations or additional text outside the JSON response.

Categories:

Prescription Medicine, Over-the-Counter (OTC), Vitamin & Supplements, Medical Equipment, Baby Care, Personal Care, Diabetes Care, First Aid, Health Monitoring Device, Others

Expected Output Format:

```
[
  {
    "purchase_date": "",
    "customer_name": "",
    "pharmacy_name": "",
    "medicine_name": [],
    "prescription_type": "",
    "bill_amount": 0,
    "payment_method": "",
    "purchase_status": "",
    "category": ""
  }
]
```

Passage:

```
{passage}
"""
```

4. Generate the structured output

```
response = client.models.generate_content(
    model='gemini-3.5-flash',
    contents=prompt,
    config=types.GenerateContentConfig(
        response_mime_type="application/json",
        temperature=0.0, # 0.0 guarantees strict, deterministic data
        extraction
    ),
)
```

5. Parse and print the resulting JSON

try:

```

extracted_data = json.loads(response.text)
print(json.dumps(extracted_data, indent=2))
except json.JSONDecodeError:
    print("Failed to decode JSON. Raw response:")
    print(response.text)

```

```

[
  {
    "purchase_date": "18 November",
    "customer_name": "Mr. Arun Kumar",
    "pharmacy_name": "MediPlus Pharmacy",
    "medicine_name": [
      "Amoxicillin 500 mg",
      "Paracetamol 650 mg"
    ],
    "prescription_type": "Prescription",
    "bill_amount": 785,
    "payment_method": "UPI",
    "purchase_status": "Completed",
    "category": "Prescription Medicine"
  },
  {
    "purchase_date": "18 November",
    "customer_name": "Mrs. Kavitha Raman",
    "pharmacy_name": "MediPlus Pharmacy",
    "medicine_name": [
      "Vitamin D3 Capsules",
      "Calcium Tablets",
      "Multivitamin Syrup"
    ],
    "prescription_type": "No Prescription",
    "bill_amount": 1280,
    "payment_method": "Credit Card",
    "purchase_status": "Completed",
    "category": "Vitamin & Supplements"
  },
  {
    "purchase_date": "18 November",
    "customer_name": "Mr. Mohammed Irfan",
    "pharmacy_name": "MediPlus Pharmacy",
    "medicine_name": [
      "Insulin Pens",
      "Blood Glucose Test Strips",
      "Lancets"
    ],
    "prescription_type": "Unknown",
    "bill_amount": 3650,
    "payment_method": "Net Banking",
    "purchase_status": "Completed",
    "category": "Diabetes Care"
  },
  {
    "purchase_date": "19 November",
    "customer_name": "Mrs. Priya Senthil",
    "pharmacy_name": "MediPlus Pharmacy",
    "medicine_name": [
      "Digital Thermometer",

```

```
    "Pulse Oximeter"
  ],
  "prescription_type": "Unknown",
  "bill_amount": 2150,
  "payment_method": "Debit Card",
  "purchase_status": "Completed",
  "category": "Health Monitoring Device"
},
{
  "purchase_date": "19 November",
  "customer_name": "Mr. Vinoth Raj",
  "pharmacy_name": "MediPlus Pharmacy",
  "medicine_name": [
    "Antiseptic Solution",
    "Cotton Rolls",
    "Sterile Gauze Pads",
    "Adhesive Bandages"
  ],
  "prescription_type": "Unknown",
  "bill_amount": 620,
  "payment_method": "Unknown",
  "purchase_status": "Completed",
  "category": "First Aid"
},
{
  "purchase_date": "20 November",
  "customer_name": "Mrs. Deepika Anand",
  "pharmacy_name": "MediPlus Pharmacy",
  "medicine_name": [
    "Baby Cough Syrup",
    "Infant Vitamin Drops",
    "Baby Skin Cream"
  ],
  "prescription_type": "Prescription",
  "bill_amount": 1420,
  "payment_method": "Google Pay",
  "purchase_status": "Completed",
  "category": "Baby Care"
},
{
  "purchase_date": "20 November",
  "customer_name": "Mr. Sanjay Verma",
  "pharmacy_name": "MediPlus Pharmacy",
  "medicine_name": [
    "Blood Pressure Tablets",
    "Cholesterol Medication",
    "Vitamin B12 Tablets"
  ],
  "prescription_type": "Unknown",
  "bill_amount": 1860,
  "payment_method": "PhonePe",
  "purchase_status": "Completed",
  "category": "Prescription Medicine"
},
{
  "purchase_date": "21 November",
  "customer_name": "Mrs. Nisha George",
```

```
"pharmacy_name": "MediPlus Pharmacy",
"medicine_name": [
    "Face Wash",
    "Moisturizer",
    "Sunscreen Lotion",
    "Hand Sanitizer"
],
"prescription_type": "Unknown",
"bill_amount": 1190,
"payment_method": "Cash",
"purchase_status": "Completed",
"category": "Personal Care"
},
{
    "purchase_date": "21 November",
    "customer_name": "Mr. Ashok Babu",
    "pharmacy_name": "MediPlus Pharmacy",
    "medicine_name": [
        "Digital Blood Pressure Monitor"
    ],
    "prescription_type": "Unknown",
    "bill_amount": 2950,
    "payment_method": "Credit Card",
    "purchase_status": "Completed",
    "category": "Health Monitoring Device"
},
{
    "purchase_date": "21 November",
    "customer_name": "Mrs. Shalini Raj",
    "pharmacy_name": "MediPlus Pharmacy",
    "medicine_name": [
        "Pain Relief Spray",
        "Muscle Relaxant Gel",
        "Pain Relief Tablets"
    ],
    "prescription_type": "Unknown",
    "bill_amount": 875,
    "payment_method": "Unknown",
    "purchase_status": "Completed",
    "category": "Over-the-Counter (OTC)"
}
]
```

In []: